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United States Patent	12399539
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Lee; Kun-Cheng

Waterproof electronic device and fan device thereof

Abstract

A waterproof electronic device and a fan device thereof are provided. The waterproof electronic device includes a housing, a system component region, a waterproof wall, and the fan device. The system component region includes a first connecting device exposed to the housing. The waterproof wall is disposed between the housing and the system component region. A fan receiving area, a wire passing area, and an opening exposed to the housing are defined by the waterproof wall. The opening corresponds in position to the first connecting device. The fan device includes a fan lid, a fan body, a cable, and a second connecting device that is coupled to the first connecting device. The fan lid is fixed to the waterproof wall and covers the fan receiving area. The cable is connected to the fan body and the second connecting device, and passes through the wire passing area.

Inventors: **Lee; Kun-Cheng (Taipei, TW)**

Applicant: **Getac Technology Corporation** (New Taipei, TW)

Family ID: **1000008782002**

Assignee: **Getac Technology Corporation (New Taipei, TW)**

Appl. No.: **18/164601**

Filed: **February 05, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20230400891 A1	Dec. 14, 2023

Foreign Application Priority Data

CN	202211461703.5	Nov. 17, 2022

Related U.S. Application Data

us-provisional-application US 63350914 20220610

Publication Classification

Int. Cl.: **H05K7/20** (20060101); **G06F1/16** (20060101); **G06F1/20** (20060101); **H01R13/52** (20060101); **H01R33/965** (20060101); H01R33/74 (20060101); H01R33/97 (20060101)

U.S. Cl.:

CPC **G06F1/1658** (20130101); **G06F1/203** (20130101); **H01R13/5208** (20130101); **H01R33/965** (20130101); **H05K7/20145** (20130101); **H05K7/20154** (20130101); **H05K7/20172** (20130101); H01R33/74 (20130101); H01R33/97 (20130101); H01R2201/06 (20130101)

Field of Classification Search

CPC: G06F (1/1658); G06F (1/203); H01R (13/5208); H01R (33/965); H01R (33/74); H01R (33/97); H01R (2201/06); H05K (7/20145); H05K (7/20154); H05K (7/20172)

USPC: 361/679.49

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7054155	12/2005	Mease et al.	N/A	N/A
7158376	12/2006	Richardson	361/679.56	G06F 1/1626
7336489	12/2007	Chen	165/104.34	G06F 1/203
8111513	12/2011	Yeh	174/16.3	G06F 1/20
8759721	12/2013	Alexander	165/61	A47J 36/321
9383788	12/2015	Reber et al.	N/A	N/A
2004/0212960	12/2003	Shih	N/A	N/A
2005/0030707	12/2004	Richardson	361/679.56	G06F 1/1626
2009/0103265	12/2008	Hongo	N/A	N/A
2010/0309623	12/2009	Yeh	361/679.48	G06F 1/203
2013/0213617	12/2012	Chou	165/121	F28F 13/12
2015/0253819	12/2014	Choi	361/679.02	G06F 1/182
2021/0059065	12/2020	Lee	N/A	H05K 5/023
2021/0089090	12/2020	Lee	N/A	E05D 3/02
2021/0408659	12/2020	Lee	N/A	G06F 1/1698
2022/0091638	12/2021	Lee	N/A	H04M 1/0274
2022/0200169	12/2021	Lee	N/A	H01Q 1/243

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
104847680	12/2017	CN	N/A
104808743	12/2018	CN	N/A

114167945	12/2021	CN	N/A
216217932	12/2021	CN	N/A
102014002452	12/2013	DE	N/A
2018206958	12/2017	JP	N/A
453470	12/2000	TW	N/A
M504440	12/2014	TW	N/A
I614592	12/2017	TW	N/A
WO2004100262	12/2003	WO	N/A

OTHER PUBLICATIONS

Office Action issued on Oct. 27, 2023 for AU application No. 2023200811. cited by applicant

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Primary Examiner: Vortman; Anatoly

Attorney, Agent or Firm: Cooper Legal Group, LLC

Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATION (1) This application claims the benefit of priority to the U.S. Provisional Patent Application Ser. No. 63/350,914 filed on Jun. 10, 2022, and to China Patent Application No. 202211461703.5 filed on Nov. 17, 2022 in People's Republic of China, both of which applications are incorporated herein by reference in their entirety. (2) Some references, which may include patents, patent applications and various publications, may be cited and discussed in the description of this disclosure. The citation and/or discussion of such references is provided merely to clarify the description of the present disclosure and is not an admission that any such reference is “prior art” to the disclosure described herein. All references cited and discussed in this specification are incorporated herein by reference in their entireties and to the same extent as if each reference was individually incorporated by reference.

FIELD OF THE DISCLOSURE
(1) The present disclosure relates to a waterproof electronic device and a fan device thereof, and more particularly to an electronic device having waterproof and dustproof functions, and a fan device installed in the electronic device.
BACKGROUND OF THE DISCLOSURE
(2) An industrial computer (or referred to as a rugged laptop) is usually used outdoors. Since an environment in which the industrial computer is used is generally poor, the industrial computer is required to have high-temperature resistant, low-temperature resistant, heat-dissipating, dustproof, and waterproof properties. In order for the industrial computer to have a high heat-dissipating effect, like traditional desktop computers or laptops, a heat-dissipating fan and heat-dissipating holes are preferably provided on components that radiate a large amount of heat (e.g., a central processing unit, a hard disk, or a memory). However, due to the poor environment, such a heat-dissipating approach may not achieve the goals of being dustproof and waterproof. On the other hand, a dustproof and waterproof structure may cause inconveniences in dismount and maintenance of the heat-dissipating fan.
(3) Therefore, how to enhance maintenance conveniences and waterproof and dustproof effects of a waterproof electronic device and a fan device thereof through an improvement in structural design, so as to overcome the above-mentioned deficiencies, has become one of the important issues to be solved in this industry.
SUMMARY OF THE DISCLOSURE

(4) In response to the above-referenced technical inadequacies, the present disclosure provides a waterproof electronic device and a fan device thereof.
(5) In order to solve the above-mentioned problems, one of the technical aspects adopted by the present disclosure is to provide a waterproof electronic device, which includes a housing, a system component region, a waterproof wall, and a fan device. The system component region includes a first connecting device exposed to the housing. The waterproof wall is disposed between the housing and the system component region. A fan receiving area, a wire passing area and an opening exposed to the housing are defined by the waterproof wall. The wire passing area is arranged between the fan receiving area and the opening, and the opening is corresponded in position to the first connecting device. The fan device includes a fan lid, a fan body, a cable and a second connecting device. The fan lid is fixed to the waterproof wall, covers the fan receiving area, and extends to the opening. The fan body is disposed in the fan receiving area. The cable is connected to the fan body and the second connecting device, and passes through the wire passing area, and the second connecting device is coupled to the first connecting device.
(6) In order to solve the above-mentioned problems, another one of the technical aspects adopted by the present disclosure is to provide a fan device. The fan device includes a fan lid, a fan body, a cable, and a fan-end connecting device. The fan body is fixed to one end of the fan lid. The fan-end connecting device is disposed at another end of the fan lid. The cable is connected to the fan body and the fan-end connecting device.
(7) Therefore, in the waterproof electronic device and the fan device thereof provided by the present disclosure, the fan device provides the fan-end connecting device (or referred to as the second connecting device) that is pluggable. The second connecting device is connected to the first connecting device in a pluggable manner, so that the fan device can be disposed in a replaceable manner in the waterproof electronic device.
(8) In addition, by virtue of "a waterproof wall being disposed between the housing and the system component region," "a fan receiving area, a wire passing area, and an opening exposed to the housing being defined by the waterproof wall," and "wherein the cable is connected to the fan body and the second connecting device and passing through the wire passing area, and the second connecting device is coupled to the first connecting device," a waterproof performance of the waterproof electronic device can be enhanced.
(9) These and other aspects of the present disclosure will become apparent from the following description of the embodiment taken in conjunction with the following drawings and their captions, although variations and modifications therein may be affected without departing from the spirit and scope of the novel concepts of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The described embodiments may be better understood by reference to the following description and the accompanying drawings, in which:
(2) FIG. **1** is a schematic perspective view of a waterproof electronic device according to the present disclosure;
(3) FIG. **2** is a schematic bottom perspective view of the waterproof electronic device according to the present disclosure;
(4) FIG. **3** is a partial schematic exploded view of the waterproof electronic device and a fan device according to the present disclosure;
(5) FIG. **4** is another partial schematic exploded view of the waterproof electronic device and the fan device according to the present disclosure;
(6) FIG. **5** is a schematic exploded view of the fan device according to the present disclosure;

(7) FIG. **6** is another schematic exploded view of the fan device according to the present disclosure;
(8) FIG. **7** is a partial cross-sectional perspective view of the waterproof electronic device according to the present disclosure; and
(9) FIG. **8** is a partial cross-sectional view of the waterproof electronic device according to the present disclosure.

DETAILED DESCRIPTION OF THE EXEMPLARY EMBODIMENTS

(10) The present disclosure is more particularly described in the following examples that are intended as illustrative only since numerous modifications and variations therein will be apparent to those skilled in the art. Like numbers in the drawings indicate like components throughout the views. As used in the description herein and throughout the claims that follow, unless the context clearly dictates otherwise, the meaning of “a”, “an”, and “the” includes plural reference, and the meaning of “in” includes “in” and “on”. Titles or subtitles can be used herein for the convenience of a reader, which shall have no influence on the scope of the present disclosure.

(11) The terms used herein generally have their ordinary meanings in the art. In the case of conflict, the present document, including any definitions given herein, will prevail. The same thing can be expressed in more than one way. Alternative language and synonyms can be used for any term(s) discussed herein, and no special significance is to be placed upon whether a term is elaborated or discussed herein. A recital of one or more synonyms does not exclude the use of other synonyms. The use of examples anywhere in this specification including examples of any terms is illustrative only, and in no way limits the scope and meaning of the present disclosure or of any exemplified term. Likewise, the present disclosure is not limited to various embodiments given herein. Numbering terms such as “first”, “second” or “third” can be used to describe various components, signals or the like, which are for distinguishing one component/signal from another one only, and are not intended to, nor should be construed to impose any substantive limitations on the components, signals or the like.

(12) Referring to FIG. **1** to FIG. **2**, the present disclosure provides a waterproof electronic device **100**. For example, said electronic device can be applied to a waterproof laptop computer, and more particularly to an industrial laptop computer. As shown in FIG. **1**, the waterproof electronic device **100** includes a device main body **101** and a display screen **102**. For convenience of illustration, a keyboard and an outer casing are not shown in FIG. **1**. The display screen **102** is rotatably connected to one side of the device main body **101**. However, the present disclosure is not limited thereto. For example, said electronic device can also be applied to a tablet computer or other electronic devices having a waterproof function (which are not necessarily equipped with a monitor).

(13) In this embodiment, an electronic component that has a good waterproof performance is disposed in the waterproof electronic device **100**, and can prevent liquid or dust from entering into the device main body **101**. More importantly, the electronic component is replaceable. Specifically, the electronic component is exemplified as a fan device at a back side of a laptop computer.

(14) Reference is made to FIG. **2** to FIG. **4**. The waterproof electronic device **100** includes a housing **20**, a system component region **20**S, a waterproof wall **20**W, and a fan device **30**. The housing **20** is hollow, and the material of the housing **20** can be metal. A keyboard (not shown) and a touch pad (not shown) can be disposed on a top surface of the housing **20**. As shown in FIG. **2**, at least one receiving area (**201**, **202**) is formed on a lower surface of the housing **20** for accommodating, for example, a battery. The system component region **20**S is formed inside the housing **20** for receiving electronic elements (e.g., a main circuit board, not shown). The waterproof wall **20**W is concaved from a back surface of the housing **20**, and is integrally formed with the housing **20**. The waterproof wall **20**W corresponds in shape to the fan device **30**. In this embodiment, the waterproof wall **20**W does not occupy a peripheral space of the device main body **101** of the laptop computer, such that other input/output ports can be flexibly arranged in the peripheral space.

(15) The system component region **20**S includes a first connecting device **21** exposed to the housing **20**. The first connecting device **21** can be referred to as a system-end connecting device. The first connecting device **21** has one end that is internally connected to the main circuit board (not shown), and has another end that is externally and electrically connected to the fan device **30**.
(16) The waterproof wall **20**W is disposed between the housing **20** and the system component region **20**S, and is concaved in the housing **20** in an integral manner. The waterproof wall **20**W is formed with a fan receiving area **23**, a wire passing area **24**, and an opening **250** exposed to the housing **20**. The wire passing area **24** is located between the fan receiving area **23** and the opening **250**. The opening **250** is corresponded in position to the first connecting device **21**. The fan receiving area **23** is used to receive the fan device **30**.
(17) Referring to FIG. **4** to FIG. **6**, the fan device **30** includes a fan lid **31**, a fan body **33**, a cable **34**, and a second connecting device **35**. The fan lid **31** is fixed to the waterproof wall **20**W, covers the fan receiving area **23**, and extends to the opening **250**. Specifically, the fan lid **31** is fastened to the waterproof wall **20**W (as shown in FIG. **2**) by fixing members S**1**, S**2**. The fixing member can be, for example, screws. In this embodiment, an airflow enters into the fan body **33** from the fan lid **31**. The fan body **33** is fixed at one end of the fan lid **31**. The cable **34** is connected to the fan body **33** and the second connecting device **35**, and passes through the wire passing area **24**. The second connecting device **35** can be referred to as a fan-end connecting device. The second connecting device **35** is disposed on another end of the fan lid **31**. Specifically, the second connecting device **35** is located on a bottom surface **31**S of the fan lid **31** and at one side of the fan body **33**. Preferably, the second connecting device **35** of the fan device **30** can be coupled to the first connecting device **21** in a sealed manner.
(18) As shown in FIG. **4**, in this embodiment, the waterproof wall **20**W is further formed with a connector receiving area **25**. The connector receiving area **25** has a cube shape and is slightly concaved, and corresponds in shape and position to the second connecting device **35** of the fan device **30**. The opening **250** is formed on a bottom of the connector receiving area **25**, so that the first connecting device **21** is exposed. The wire passing area **24** protrudes slightly between the fan receiving area **23** and the connector receiving area **25**. The wire passing area **24** is formed with a surface **243**. The surface **243** is lower than an outer surface **20**S of the waterproof wall **20**W, and can receive the cable **34** of the fan device **30**. The surface **243** can be a curved surface or a polygon surface. In other words, in this embodiment, the cable **34** is disposed above the surface **243** of the wire passing area **24** and is connected to the fan device **30**. The cable **34** does not penetrate through the waterproof wall **20**W.
(19) The housing **20** has a side wall **27**. The side wall **27** is formed with a fan vent **270**. An air-outlet direction F of the fan device **30** is toward the fan vent **270**. The waterproof electronic device **100** further includes a heat-dissipating fin assembly **50**. The heat-dissipating fin assembly **50** is disposed between the fan body **33** and the fan vent **270**. The heat-dissipating fin assembly **50** is disposed under the waterproof wall **20**W.
(20) As shown in FIG. **1**, a heat pipe **70** of the waterproof electronic device **100** is disposed on a top surface of the housing **20**, and is opposite to the fan device **30**. One end of the heat pipe **70** contacts a heat-generating element (which is usually a central processing unit, not shown). Another end of the heat pipe **70**, which is a heat-dissipating end, extends to contact the heat-dissipating fin assembly **50** for outward dissipation of the redundant heat. The heat pipe **70** is separated from the fan device **30**, such that the waterproof performance can be improved without affecting dismount and assembly of the fan device **30**.
(21) Referring to FIG. **4** and FIG. **5**, the fan lid **31** includes a main cover **311**, and an embedded board **312**. In this embodiment, the embedded board **312** is slightly lower than the main cover **311** and parallel to the main cover **311**. The main cover **311** is formed with an air intake **310**, and the air intake **310** corresponds in position to the fan body **33**. The embedded board **312** is extended outward from a bottom of one side **313** of the main cover **311**. An outer surface **311**S of the main

cover **311** is flush with the outer surface **20**S of the waterproof wall **20**W. The embedded board **312** is lower than the main cover **311**, and is inserted between the waterproof wall **20**W and the heat-dissipating fin assembly **50** (as shown in FIG. **7** and FIG. **8**). The embedded board **312** can be fastened to the housing **20** by the screw S**2** (as shown in FIG. **3** and FIG. **4**).
(22) Referring to FIG. **5**, in this embodiment, the fan lid **31** further includes a protective shell **315**. The protective shell **315** is formed on the bottom surface **31**S of the fan lid **31** and located at one side of the fan body **33**. As shown in FIG. **5**, the protective shell **315** has a rectangular shape formed by four walls and is not completely closed, and an opening thereof is away from the fan lid **31**. The second connecting device **35** is received in the protective shell **315**. The protective shell **315** is oriented toward the first connecting device **21** (as shown in FIG. **4**). Appearance-wise, the protective shell **315** of the fan lid **31** corresponds to the connector receiving area **25** of the waterproof wall **20**W.
(23) One side of the protective shell **315** has a gasket hole **3150** and a cable waterproof ring **3152**. The cable waterproof ring **3152** is disposed around the gasket hole **3150**. The cable **34** of the fan device **30** passes through a cable hole **3153** of the cable waterproof ring **3152**. The function of the cable waterproof ring **3152** is that, even if liquid or dust accidentally enters the fan lid **31**, the cable waterproof ring **3152** can block the liquid or the dust from entering the protective shell **315** (i.e., from entering the second connecting device **35** of the fan device **30**).
(24) Referring to FIG. **4**, FIG. **7** and FIG. **8**, in this embodiment, the first connecting device **21** has a plurality of elastic contacts **211**. The elastic contacts **211** can be, for example, pogo pins. However, the present disclosure is not limited thereto. The elastic contacts **211** are oriented toward the second connecting device **35**. Correspondingly, the second connecting device **35** includes a circuit board **351**, and the circuit board **351** is fastened in the protective shell **315** by a screw (not shown). One surface **350** of the circuit board **351** has a plurality of contact pads **352** formed thereon. The elastic contacts **211** correspondingly contact the contact pads **352**.
(25) Referring to FIG. **5** and FIG. **6**, the second connecting device **35** further includes a cable connector **353**. The cable connector **353** is disposed on another surface **353** of the circuit board **351**, which is opposite to the contact pads **352**. The cable connector **353** and the contact pads **352** are disposed on two opposite sides of the circuit board **351**. The cable **34** is connected to the cable connector **353**, and is externally connected to the fan device **30**. An electrical connection sequence is arranged in the following order. The contact pads **352** are connected to the cable connector **353** through the circuit board **351**. Then, the cable connector **353** is connected to the fan body **33** of the fan device **30** by the cable **34**. The cable **34** passes through the cable waterproof ring **3152**, and penetrates through the gasket hole **3150** of the protective shell **315**.
(26) Referring to FIG. **7** and FIG. **8**, the fan device **30** can further include a first waterproof ring R**1**. The first waterproof ring R**1** is disposed on a periphery of the fan body **33**. Specifically, the first waterproof ring R**1** can be configured to surround a periphery of the protective shell **315**, a periphery of an inner surface of the main cover **311**, and a corner between the main cover **311** and the embedded board **312**. The first waterproof ring R**1** can be strip-shaped and closed for serving as a first waterproof line, so as to enhance the waterproof performance of this embodiment.
(27) In addition, the fan device **30** further includes a second waterproof ring R**2**. The second waterproof ring R**2** is disposed between the first connecting device **21** and the second connecting device **35**. Specifically, the second waterproof ring R**2** is disposed on a bottom surface of the connector receiving area **25**, and surrounds the opening **250**. An edge of the protective shell **315** is in contact with the second waterproof ring R**2**. The second waterproof ring R**2** can be formed to have a flat and rectangular shape, so that the second connecting device **35** can be coupled to the first connecting device **21** in a sealed manner.
(28) As shown in FIG. **7** and FIG. **8**, there are three protective lines between the second connecting device **35** and the first connecting device **21**, which are the first waterproof ring R**1**, the second waterproof ring R**2**, and the cable waterproof ring **3152**. In particular, the second waterproof ring

R**2** is disposed between the edge of the protective shell **315** and the bottom surface of the connector receiving area **25**, and can provide a good waterproof function. In addition to a good waterproof function between the second connecting device **35** and the first connecting device **21** of the present disclosure, the fan device **30** can be replaceable in this embodiment due to the second connecting device **35** being detachably connected to the first connecting device **21**. If the fan device **30** malfunctions, a user can easily dismount the fan device **30** from the waterproof wall **20**W of the waterproof electronic device **100** by removing screws (S**1**, S**2**, as shown in FIG. **2**) on the fan lid **31**. Maintenance of the fan device **30** of this embodiment is convenient.

Beneficial Effects of the Embodiment

(29) In conclusion, at least one of the effects in the waterproof electronic device and the fan device thereof provided by the present disclosure, the fan device **30** provides the fan-end connecting device (or referred to as the second connecting device **35**) that is pluggable. The second connecting device **35** is connected to the first connecting device **21** in a pluggable manner, so that the fan device **30** can be disposed in a replaceable manner in the waterproof electronic device **100**.

(30) In addition, by virtue of "a waterproof wall being disposed between the housing and the system component region," "a fan receiving area, a wire passing area, and an opening exposed to the housing being defined by the waterproof wall," and "the cable being connected to the fan body and the second connecting device and passing through the wire passing area, and the second connecting device being coupled to the first connecting device," the waterproof performance of the waterproof electronic device can be enhanced.

(31) The foregoing description of the exemplary embodiments of the disclosure has been presented only for the purposes of illustration and description and is not intended to be exhaustive or to limit the disclosure to the precise forms disclosed. Many modifications and variations are possible in light of the above teaching.

(32) The embodiments were chosen and described in order to explain the principles of the disclosure and their practical application so as to enable others skilled in the art to utilize the disclosure and various embodiments and with various modifications as are suited to the particular use contemplated. Alternative embodiments will become apparent to those skilled in the art to which the present disclosure pertains without departing from its spirit and scope.

Claims

1. A waterproof electronic device, comprising: a housing; a system component region, comprising a first connecting device exposed to the housing; a waterproof wall, wherein a fan receiving area, a wire passing area, and an opening exposed to the housing are defined by the waterproof wall, the wire passing area is arranged between the fan receiving area and the opening the opening corresponds in position to the first connecting device, and the first connecting device comprises a plurality of contacts exposed through the opening; and a fan device, comprising a fan lid, a fan body, a cable, and a second connecting device, wherein the fan lid is fixed to the waterproof wall, covers the fan receiving area, and extends to the opening, the fan body is disposed in the fan receiving area, the cable is connected to the fan body and the second connecting device and passes through the wire passing area, and the second connecting device is coupled to the first connecting device.

2. The waterproof electronic device according to claim 1, wherein the fan body is fixed to the fan lid.

3. The waterproof electronic device according to claim 1, wherein a connector receiving area is further defined by the waterproof wall, the opening is formed at a bottom of the connector receiving area, the wire passing area protrudes between the fan receiving area and the connector receiving area, the wire passing area is formed to have a concave surface, and the concave surface is lower than an outer surface of the waterproof wall.

4. The waterproof electronic device according to claim 1, wherein the housing has a side wall, a fan vent is formed on the side wall, and an air-outlet direction of the fan device is toward the fan vent.
5. The waterproof electronic device according to claim 4, further comprising a heat-dissipating fin assembly, wherein the heat-dissipating fin assembly is disposed between the fan body and the fan vent, and the heat-dissipating fin assembly is disposed under the waterproof wall.
6. The waterproof electronic device according to claim 5, wherein the fan lid comprises a main cover and an embedded board, an air intake is formed on the main cover, the air intake corresponds in position to the fan body, the embedded board extends outward from a bottom of one side of the main cover, an outer surface of the main cover is flush with an outer surface of the waterproof wall, the embedded board is lower than the main cover, and the embedded board is inserted between the waterproof wall and the heat-dissipating fin assembly.
7. The waterproof electronic device according to claim 1, wherein the fan lid further includes a protective shell, the protective shell is formed on a bottom surface of the fan lid, the second connecting device is received in the protective shell, the protective shell is oriented toward the first connecting device, the protective shell has a gasket hole and a cable waterproof ring, the cable waterproof ring is disposed on the gasket hole, and the cable passes through the cable waterproof ring.
8. The waterproof electronic device according to claim 1, wherein the plurality of contacts are a plurality of elastic contacts, the plurality of elastic contacts are oriented toward the second connecting device, the second connecting device comprises a circuit board, a plurality of contact pads are disposed on one surface of the circuit board, and the plurality of elastic contacts correspondingly contact the plurality of contact pads.
9. The waterproof electronic device according to claim 8, wherein the second connecting device further comprises a cable connector, the cable connector is disposed on another surface of the circuit board, and the cable is connected to the cable connector.
10. The waterproof electronic device according to claim 1, wherein the fan device further comprises a first waterproof ring, and the first waterproof ring is configured to surround a periphery of the fan body.
11. The waterproof electronic device according to claim 10, wherein the fan device further comprises a second waterproof ring, and the second waterproof ring is disposed between the first connecting device and the second connecting device.
12. The waterproof electronic device according to claim 1, wherein the wire passing area is between the fan receiving area and the opening.
13. The waterproof electronic device according to claim 1, wherein the second connecting device comprises a plurality of contact pads correspondingly contacting the plurality of contacts.